

2.3 Linear Equations

A first order Linear DE has the form:

$$a_1(x) \frac{dy}{dx} + a_0(x) y = g(x).$$

By dividing both sides by $a_1(x)$, we obtain the standard form of a linear DE.

$$\boxed{\frac{dy}{dx} + P(x) y = f(x)} \quad \leftarrow \begin{array}{l} \text{standard} \\ \text{form} \end{array}$$

To solve the standard form DE, we multiply both sides by an integrating

Factor:

$$\boxed{e^{\int P(x) dx}} \quad \leftarrow \begin{array}{l} \text{integrating} \\ \text{factor} \end{array}$$

$$\Rightarrow e^{\int P(x) dx} \frac{dy}{dx} + e^{\int P(x) dx} P(x) y = f(x) e^{\int P(x) dx}$$

The left side corresponds to
a single derivative of a
Product.

$$\Rightarrow \frac{d}{dx} \left[y e^{\int P(x) dx} \right] = f(x) e^{\int P(x) dx}$$

Next, we can integrate both sides
and solve for y .

Note: The solution obtained is valid
inside intervals where both $P(x)$ and
 $f(x)$ are continuous.

To solve a first order linear DE

1. write the equation in standard form

$$\frac{dy}{dx} + P(x)y = f(x)$$

2. Multiply both sides by "the integrating factor"

$$e^{\int P(x) dx}$$

3. The left hand side becomes

$$\frac{d}{dx} \left[y e^{\int P(x) dx} \right]$$

4. Integrate both sides with respect to x
and solve for y .

ex: ① solve the DE

$$2 \frac{dy}{dx} + 4y = 5.$$

It is a Linear DE. Its standard form is

$$\frac{dy}{dx} + \underset{\substack{\uparrow \\ P(x)}}{2} y = \frac{5}{2} \quad (*)$$

$$\begin{aligned} \text{Integrating factor: } e^{\int P(x) dx} &= e^{\int 2 dx} \\ &= e^{2x} \end{aligned}$$

Multiply (*) by e^{2x} to get:

$$e^{2x} \frac{dy}{dx} + 2e^{2x} y = \frac{5}{2} e^{2x}$$

$$\Rightarrow \frac{d}{dx} [y e^{2x}] = \frac{5}{2} e^{2x}$$

Integrate both sides

$$\int \frac{d}{dx} [y e^{ax}] dx = \int \frac{5}{2} e^{ax} dx$$

$$\Rightarrow y e^{ax} = \frac{5}{4} e^{ax} + C$$

$$\Rightarrow \boxed{y = \frac{5}{4} + C e^{-ax}}$$

since $P(x) = 2$ and $f(x) = 5/2$

are continuous for all x , the solution is valid for all x in $(-\infty, \infty)$.

A useful Log Property
to simplify integrating
factors is :

$$e^{n \ln(x)} = e^{\ln(x^n)} = x^n$$

For example :

$$e^{3 \ln x} = e^{\ln(x^3)} = x^3$$

$$e^{-\ln x} = e^{\ln(x^{-1})} = x^{-1}$$

ex: ② Solve the IVP:

$$\begin{cases} x^3 y' + 4x^2 y = e^x \\ y(1) = 2 \end{cases}$$

The DE is Linear. Its standard form is

$$y' + \underbrace{\left(\frac{4}{x}\right)}_{P(x)} y = \frac{e^x}{x^3} \quad (*)$$

$$\begin{aligned} \text{I.F. } e^{\int \frac{4}{x} dx} &= e^{4 \ln|x|} = e^{\ln|x^4|} = |x^4| \\ &= x^4 \end{aligned}$$

Multiply (*) by x^4 to get:

$$x^4 y' + 4x^3 y = x e^x$$

$$\Rightarrow (x^4 y)' = x e^x$$

Integrate both sides to get:

$$x^4 y = \int x e^x dx$$

D	I
x	e^x
1	e^x
0	e^x

$$x^4 y = x e^x - e^x + C$$

$$\Rightarrow y = \frac{x e^x - e^x + C}{x^4}$$

Let's find C from $y(1) = 2$

set $x=1$ and $y=2$.

$$\Rightarrow 2 = \frac{e^1 - e^1 + C}{1} \Rightarrow C = 2$$

Then :

$$y = \frac{x e^x - e^x + 2}{x^4}$$

Since $P(x) = \frac{4}{x}$ and $f(x) = \frac{e^x}{x^3}$

are not continuous at $x=0$, the
general solution of the DE (*) is
valid for $(-\infty, 0)$ or $(0, \infty)$.

For the IVP, the initial condition is
 $y(1) = 2$ so the interval needs to
include $x=1$.

Then, the solution of the IVP

that we found: $y = \frac{xe^x - e^x + 2}{x^4}$

is valid over the interval: $(0, \infty)$.

Integrating Factor (I.F.)

When you find an integrating factor

with $e^{\int P(x) dx}$, you don't need

a constant of integration for $\int P(x) dx$.

$$e^{\int P(x) dx} + C = e^C e^{\int P(x) dx}$$

The e^C would cancel out once we multiply both sides of a DE by the I.F.

We don't have to worry about absolute value when we find I.F.

e.g. If $e^{\int P(x) dx} = e^{\int \frac{3}{x} dx} = e^{3 \ln|x|}$
 $= e^{\ln|x^3|} = |x^3|$

We can use x^3 as the I.F.

ex: ③ Solve the IVP $\begin{cases} \frac{dx}{dt} + 2x = 4t \\ x(0) = 5 \end{cases}$

The DE is Linear in standard form.

$$P(t) = 2$$

$$\text{I.F. } e^{\int P(t) dt} = e^{2t}$$

$$\Rightarrow e^{2t} \frac{dx}{dt} + 2x e^{2t} = 4t e^{2t}$$

$$\Rightarrow \frac{d}{dt} [x e^{2t}] = 4t e^{2t}$$

$$\begin{aligned} \Rightarrow x e^{2t} &= \int 4t e^{2t} dt \\ &= 2t e^{2t} - e^{2t} + C \end{aligned}$$

$$\Rightarrow x = 2t - 1 + C e^{-at}$$

Let's find C from $x(0) = 5$

set $t=0$ and $x=5$

$$\Rightarrow 5 = 0 - 1 + C e^0$$

$$\Rightarrow C = 6$$

The IVP solution is :

$$x = 2t - 1 + 6 e^{-at}$$

Note : As $t \rightarrow \infty$, $6 e^{-at} \rightarrow 0$

The term $6 e^{-at}$ is a Transient term.

ex: ④ [2.3 #15] solve the DE

$$y dx - 4(x + y^6) dy = 0$$

The DE is equivalent to : $y dx = 4(x + y^6) dy$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{4(x + y^6)}$$

This DE is not separable and is not linear in y .

Instead, think of x as a function of y .

$$\Rightarrow \frac{dx}{dy} = \frac{4(x + y^6)}{y} = \frac{4x}{y} + 4y^5$$

$$\Rightarrow \frac{dx}{dy} - \frac{4}{y} x = 4y^5$$

(The term $\frac{4}{y} x$ is circled with a dashed line, and an arrow points from the label $P(y)$ to it.)

This is a Linear DE in x .

I.F. $e^{\int P(y) dy} = e^{\int -\frac{4}{y} dy}$

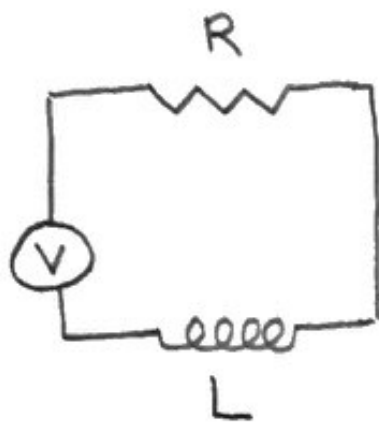
$$= e^{-4 \ln y}$$
$$= e^{\ln(y^{-4})}$$
$$= y^{-4}$$

$$\Rightarrow \frac{d}{dy} [x y^{-4}] = 4y$$

$$\Rightarrow x y^{-4} = \int 4y dy = 2y^2 + C$$

$$\Rightarrow \boxed{x = 2y^6 + C y^4}$$

ex: ⑤ Consider a circuit with
a resistance R , an inductor L ,
and a voltage source V .



Assume L , R , and V
are constants.

Find the current $i(t)$ in the circuit
if the initial current (at $t=0$) is i_0 .

The DE for the current is :

$$L \frac{di}{dt} + Ri = V$$

The DE is Linear. The standard form is:

$$\frac{di}{dt} + \underbrace{\frac{R}{L}}_{P(t)} i = \frac{V}{L} \quad (*)$$

I.F. $e^{\int P(t) dt} = e^{\int \frac{R}{L} dt} = e^{\frac{Rt}{L}}$

$$\Rightarrow e^{\frac{Rt}{L}} \frac{di}{dt} + e^{\frac{Rt}{L}} \frac{R}{L} i = \frac{V}{L} e^{\frac{Rt}{L}}$$

$$\Rightarrow \frac{d}{dt} \left[e^{\frac{Rt}{L}} i \right] = \frac{V}{L} e^{\frac{Rt}{L}}$$

$$\Rightarrow e^{\frac{Rt}{L}} i = \int \frac{V}{L} e^{\frac{Rt}{L}} dt$$

$$\Rightarrow e^{\frac{Rt}{L}} i = \frac{V}{L} \left[\frac{e^{\frac{Rt}{L}}}{(R/L)} \right] + C$$

$$\Rightarrow e^{\frac{Rt}{L}} i = \frac{V}{R} e^{\frac{Rt}{L}} + C$$

$$\Rightarrow i = \frac{V}{R} + C e^{-Rt/L}$$

$$@ t=0, i=i_0$$

$$\Rightarrow i_0 = \frac{V}{R} + C$$

$$\Rightarrow C = i_0 - \frac{V}{R}$$

The current as a function of time t is :

$$i = \frac{V}{R} + \left(i_0 - \frac{V}{R} \right) e^{-Rt/L}$$

Note : $\lim_{t \rightarrow \infty} i = \frac{V}{R}$
